

MindCapsule — Hug the Hedgehog

Project Report (English Version)

1. Project Background and Objectives

Adolescents often experience academic stress and emotional pressure but may hesitate to seek help because of social pressure, fear of judgment, or emotional stigma. This project aims to build a privacy-focused emotional expression and support system for teenagers.

By creating a safe and non-judgmental environment, the system lowers the barrier for emotional expression, helps users recognize and regulate emotions, and promotes healthier psychological awareness.

2. Core Functions

1. Text-based Interaction and Emotion Detection:

The system analyzes users' typed messages, identifies emotional keywords and patterns, and detects possible anxiety-related emotions.

2. Emotional Destigmatization:

The system reframes negative emotions as normal psychological responses, helping users reduce shame and emotional suppression.

3. Personalized Psychological Support:

Based on emotion classification results, the system provides customized positive psychology messages and behavioral suggestions.

4. Self-guided Emotional Intervention:

The system provides a complete process including emotion recognition, acceptance, and rational coping strategies, enabling private and low-cost emotional self-support.

3. Technical Architecture

Frontend:

The interface is developed using Tkinter, providing text input, psychological prompts, and behavioral guidance components. The design uses a simple layout and low-saturation colors to reduce visual pressure and provide immediate feedback.

Backend:

A lightweight BERT-based emotion recognition model is integrated to classify emotions related to academic stress. The psychological intervention module applies cognitive restructuring and generates personalized action recommendations.

Privacy Protection:

All computation is performed locally without network transmission, ensuring that user emotional data remains private.

Local Database:

Python dictionaries are used to store psychological support messages and frequently occurring emotional keywords. Temporary data is automatically cleared after the program closes.

4. Technical Features

The system is designed for local deployment without dependence on external servers.

The optimized core algorithm achieves a response delay of less than 500ms, providing a real-time interactive experience.

5. Application Scenarios

Academic Stress Support:

When students experience pressure during late-night studying, they can communicate their feelings through text input. The system identifies anxiety-related emotions and provides supportive messages and coping suggestions.

Daily Emotional Support:

When students experience self-doubt caused by comparison with peers, the system provides positive psychological feedback and encourages healthier self-understanding.

School Psychological Assistance:

The system can support school counseling environments by helping students record emotional changes privately and providing insights into common emotional challenges.

6. Code Structure and Workflow

The project consists of a main program file and data storage module. The main program handles user interaction, emotion analysis, and response generation, while the data module stores psychological support resources.

Workflow:

1. Students enter their feelings through text input.
2. The system analyzes emotional patterns.
3. The system displays personalized positive psychological guidance.
4. Users receive emotional support and recommended coping strategies.